

Table 1: List of major ML tools and software libraries

CNTK	Apache SystemML	Caffe
Deeplearning4j	ELKI	GNU Octave
H2O	Keras	KNIME
Mahout	Mallet	Mlpack
MXNet	OpenNN	Orange
PyTorch	RapidMiner	scikit-learn
Shogun	Spark MLlib	TensorFlow
Theano	Weka	Yooreeka

Predictions using machine learning algorithms in Weka

Let us look at an example of a classifier based machine learning algorithm being used in Weka. For this, the classification problem of the data set of students is used. In this example, the data set *Students.arff* is used for training the classifier model. The marks of students are given in three phases. On the basis of marks obtained in sequence, the final class attribute is determined. In the first record, if a student scored 90, 89 and 89 marks, respectively, then the student was given admission to Stream 1. In the third record, Stream 2 is allocated to the students who obtained marks 78, 67 and 78, respectively.

```
Students.arff
@relation students
@attribute marks1 numeric
@attribute marks2 numeric
@attribute marks3 numeric
@attribute class {1, 2}
@data
90, 89, 89, 1
89, 90, 99, 1
78, 67, 78, 2
67, 71, 78, 2
69, 78, 78, 2
60, 79, 78, 2
```

The test data set *teststudents.arff* is used to predict or determine the class or stream of the students who obtained marks in a specific sequence. This problem is solved using Weka with the integration of the machine learning algorithm of J48 classifier. In the following data set, we have to determine the classes (streams) of the students on the basis of their performance (scores) in the examination.

```
teststudents.arff
@relation students
@attribute marks1 numeric
@attribute marks2 numeric
@attribute marks3 numeric
@attribute class {1, 2}
```

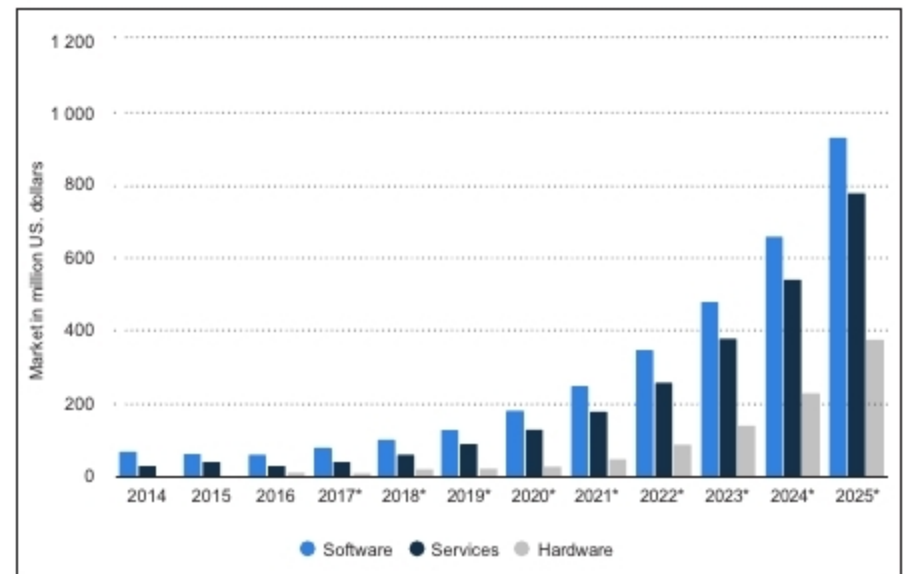


Figure 1: Predicted revenues from the US deep learning market



Figure 2: Weka Explorer

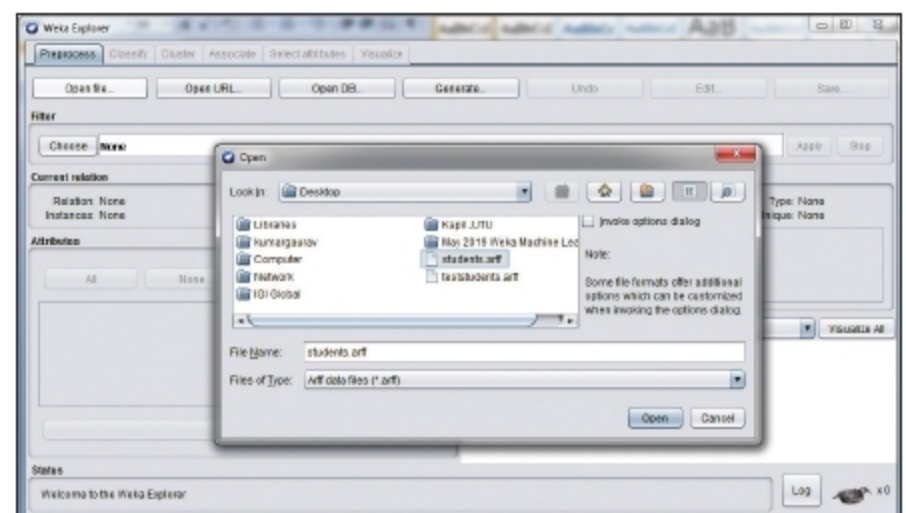


Figure 3: Reading the training data set for machine learning

```
@data
99, 91, 90, ?
89, 67, 78, ?
78, 67, 78, ?
77, 71, 78, ?
90, 78, 78, ?
10, 10, 10, ?
40, 40, 78, ?
30, 30, 80, ?
98, 97, 94, ?
```